ARP Global Capital

Technical Assessment – Local AI Investment Intelligence Platform

Company Website: https://arpglobalcapital.com/

This technical assessment evaluates practical engineering thinking across AI orchestration, DevOps, security, dashboards, infrastructure, and operational maturity.

The objective is to build a small locally runnable AI-powered investment operations platform demonstrating secure AI workflows, dashboards, RBAC, auditability, and practical DevOps workflows.

Assessment Goal

• Load investment and operational data

• Store data in a local database

• Visualize data using dashboards/charts

• Create AI agents that answer questions using real data

• Enforce role-based access control

• Maintain audit logs for AI interactions

• Run the entire platform locally using Docker

Important Notes

• The platform must run on a normal developer laptop.

• No enterprise infrastructure or GPU is required.

• No paid subscriptions or paid APIs should be required.

• The focus is architecture thinking and practical implementation.

• This is NOT a machine learning model training assessment.

Expected Timeline

Expected completion time: 3–5 days

1. Data Requirements

Candidate may either generate mock data or use free public APIs.

Option A – Generate Mock Data

• CSV files

• JSON files

• Python scripts

• Faker library

Option B – Use Free Public APIs (Optional)

• CoinGecko API

• Yahoo Finance (yfinance)

• Alpha Vantage free tier

Suggested data types include portfolio holdings, trades, market prices, users, and risk rules.

2. Database Layer

Candidate may use SQLite (recommended) or PostgreSQL.

• portfolio_holdings

• trades

• market_prices

• users

• risk_rules

• audit_logs

3. Dashboard & Visualization

The platform must include a dashboard or visualization interface.

• Streamlit

• Metabase

• Dash/Plotly

• Portfolio allocation chart

• Recent trades table

• Risk alerts table

• Asset exposure chart

4. AI Agent Requirements

The platform should include AI agents capable of answering questions using real data from the database.

Portfolio Analyst Agent

• What are our top holdings?

• What is our asset allocation?

• Which assets moved the most today?

• Summarize portfolio performance

Risk & Compliance Agent

• Are we overexposed to any asset?

• Which trades are high risk?

• Which trades need review?

• Explain why a trade was flagged

5. LLM / AI Model Requirement

• Local LLM using Ollama (bonus)

• Small open-source models

• Free external APIs

• Mock AI orchestration logic

The assessment focuses more on orchestration and tool calling rather than model quality.

6. Tool Calling / AI Orchestration

- get_portfolio_summary()
- get_recent_trades()
- get_asset_exposure()
- get_risk_alerts()
- check_user_permission()

Expected flow: User → AI Agent → Tool/API → Database → Response

7. Role-Based Access Control (Mandatory)

- analyst@local → portfolio + market data
- risk@local → portfolio + trades + risk alerts
- manager@local → summary-only access
- intern@local → limited access

The AI should return different responses depending on user role and permissions.

8. Audit Logs (Mandatory)

- user
- role
- question
- tool_called
- allowed_or_denied
- timestamp

9. Security Requirements

- No secrets committed to repository
- .env.example included

• Basic authentication/token logic

• Input validation

• RBAC enforcement

• Audit logging

Candidate should also explain risks of external LLM providers and how sensitive data leakage would be prevented.

10. DevOps & Repository Requirements

• GitHub repository

• README.md

• Architecture explanation

• Dockerfile

• docker-compose.yml

• .env.example

• Setup instructions

• Run instructions

Expected startup command: docker compose up

Optional Bonus

• GitHub Actions pipeline

• Container build pipeline

• Basic automated tests

• Better UI/UX

• RAG/vector database

11. Live Demo Requirement

• Run the platform locally during interview

• Show dashboard and charts

- Ask AI agents operational questions
- Show role-based access restrictions
- Show audit logs
- Explain architecture and technical decisions

Candidate should be prepared to explain how they would scale the platform for production use.

12. Evaluation Criteria

- Architecture thinking
- AI orchestration understanding
- DevOps maturity
- Security mindset
- Documentation quality
- Operational thinking
- Communication ability

Final Notes

Candidates are encouraged to prioritize clean implementation, practical architecture, security awareness, and clear documentation over excessive feature complexity.

The objective of this assessment is to understand how you approach system design, AI workflows, operational thinking, and engineering tradeoffs in a real-world environment.